Technical Research Report

Power Electronics and Control Systems

Single Phase Inverter:

analysis of architecture and control strategies.

Single phase inverter: investigation on different control architecture and impact on performance.

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Abstract: Single phase inverter: investigation on different control architecture and impact on performance.

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Framework: MATLAB/Simulink/Simscape

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1 Introduction

In this document a study for a single phase inverter is shown.

The document starts with a description of the hardware layout following with a description of the control architecture.

2 Architecture

2.1 Hardware Layout

System architecture can be assumed as Figure 1 shows.

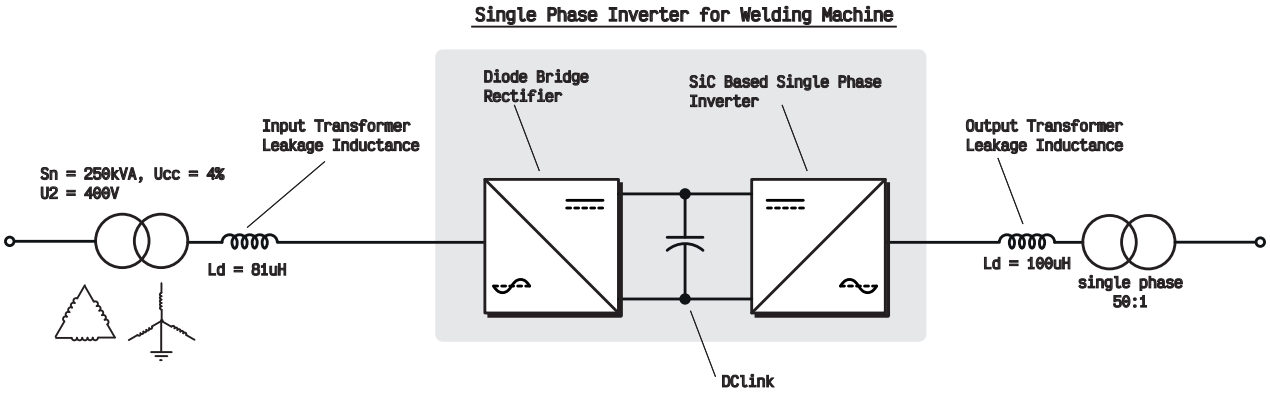


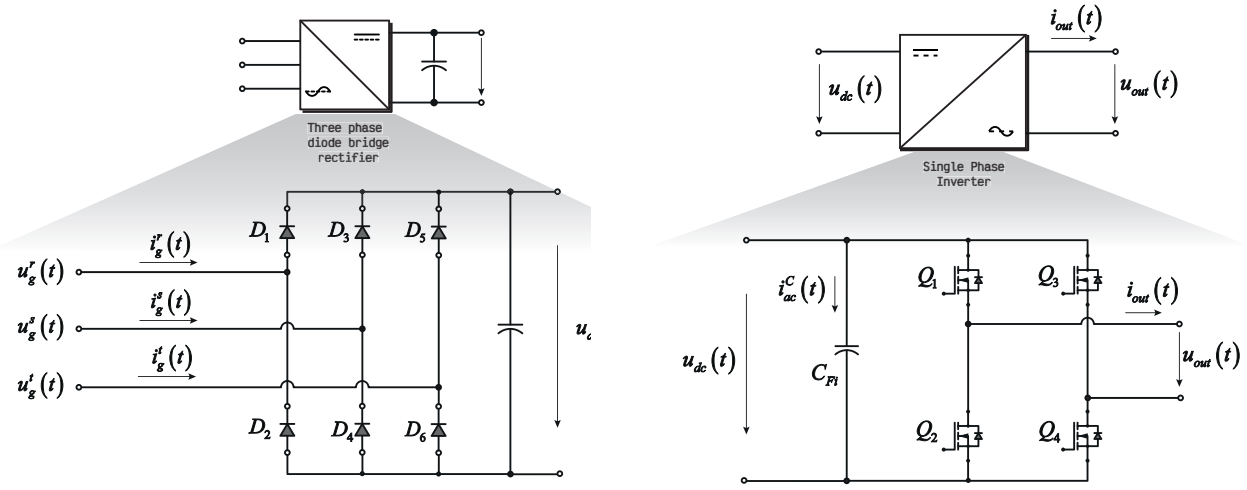
Figure 1: System layout composed by a grid with finite short circuit capability, a diode bridge rectifier, a dclink, a single phase inverter, and an output transformer with given leakage inductance.

The system is composed by the following parts:

- a grid power supply. It has been assumed that the system is connected to a 400V-250kVA three phase with a $U_{cc} = 4\,\%$ which means an equivalent leakage inductance of $L_d = 81\,\mu\text{H}$;
- a three phase diode bridge rectifier. The DClink voltage is generated by rectifying the 400V three phase voltage leading a nominal DClink voltage of $U_{dc}^{nom} = 540\,\text{V}$;
- a DClink capacitor of value $C_{Fi} = 5.6\,\text{mF}$;
- a SiC-MOSFET based full-bridge. The component selected is the module **SKM1700MB20R4S2I** operating at the switching frequency of $f_{pwm} = 20\,\text{kHz}$ per leg with an equivalent load switching frequency of $f_{load} = 40\,\text{kHz}$;

– the nominal inverter output current is $I_{out}^{nom} = 600\,\text{Arms}$;

– the load consists of an single phase transformer with a ratio factor of 50:1 (nominal secondary current of $I_2 = 30\,\text{kA}$);



(a) Input stage based on three phase diode rectifier and DClink capacitor.

(b) Single phase inverter based on SiC-MOSFET based full bridge.

Figure 2: System architecture elements.

3 Control Architectures

As mentioned, the single phase inverter will operate in current control mode. The control strategy adopted is based on a virtual DQ control, as shown in Figure 3.

The current control is implemented using a virtual DQ control (here $\xi\eta$ notation is used), where the η component is created by a phase shift filter of the form

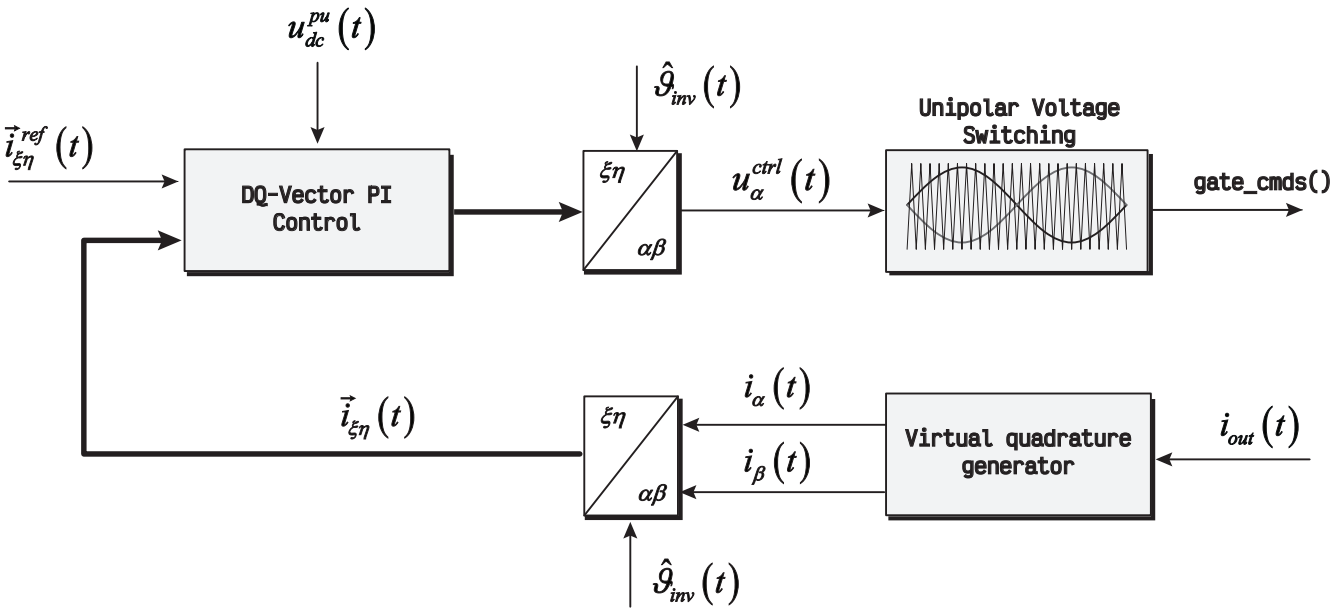


Figure 3: Current vector control layout of the single phase inverter. Virtual quadrature current is created by a phase shift filter.

3.1 Nomenclature

Here, a list of used symbols:

- $i_{out}(t)$: inverter output current;
- $u_{out}(t)$: inverter output voltage;
- $u_{dc}(t)$: DClink voltage;
- $i_{ac}^{C}(t)$: DClink capacitor current;
- I_{ac}^{b}: normalization AC current factor (end scale);
- U_{ac}^{b}: normalization AC voltage factor (end scale);
- U_{dc}^{b}: normalization DC voltage factor (end scale);
- $i_{out}^{pu}(t) = i_{out}(t)/I_{ac}^{b}$: normalized inverter output voltage;
- $i_{\alpha} = i_{out}^{pu}(t)$, load current respect to the stationary reference frame;
- $i_{\beta} = i_{\alpha}(\omega t - \pi/2)$, phase shifted load current respect to the stationary reference frame, the phase shift is generated by an *phase-shift-filter*;
- $\vartheta_{inv}(t)$: inverter output phase internally generated when $f_{inv} \neq 50\,\mathrm{Hz}$ otherwise taken from grid phase lock loop (PLL);



- $\omega = \dot{\vartheta}_{inv}(t)$
- $u_{dc}^{pu} = u_{dc}(t)/U_{dc}^{b}$: normalized DClink voltage;
- $i_{\xi}(t) = i_{\alpha}(t)\cos(\omega t) + i_{\beta}(t)\sin(\omega t)$, load current component ξ respect to the rotating reference frame;
- $i_{\eta}(t) = -i_{\alpha}(t)\sin(\omega t) + i_{\beta}(t)\cos(\omega t)$, load current component η respect to the rotating reference frame;
- $u_{\xi}^{ctrl}(t)$, output current control of component ξ respect to the rotating reference frame;
- $u_{\eta}^{ctrl}(t)$, output current control of component η respect to the rotating reference frame;
- $u_{\alpha}^{ctrl}(t) = u_{\xi}^{ctrl}(t)\cos(\omega t) - u_{\eta}^{ctrl}(t)\sin(\omega t)$, output current control respect to the stationary reference frame;

3.2 Normalization

The per unit representation or *normalization* plays a fundamental role for overall control tracking performance and shall be implemented as follows:

- Reference quantities
  - $i_b = I_{ac}^{b}$: normalization AC current factor (end scale);
  - $u_b = U_{ac}^{b}$: normalization AC voltage factor (end scale);
  - $\omega^b = 2\pi f = \frac{d}{dt}\vartheta_{inv}(t)$: normalization factor of the electrical pulsation;
- Per unit quantities
  - $R_{load}^{norm} = R_{load}/X^b$: per unit representation of the load resistance;
  - $L_{load}^{norm} = L_{load}/L^b$: per unit representation of the load inductance;
  - $\omega^{norm} = \omega/\omega^b$: per unit of the electrical pulsation;
  - $i^{norm} = i/i_b$: per unit of i current;
  - $u^{norm} = u/u_b$: per unit of u voltage;
  - $X^b = u_b/i_b$: per unit of the linkage permanent magnet flux;
  - $L^b = X^b/\omega^b$: per unit of the linkage permanent magnet flux.

3.3 Inner current control based on DQ-Vector PI

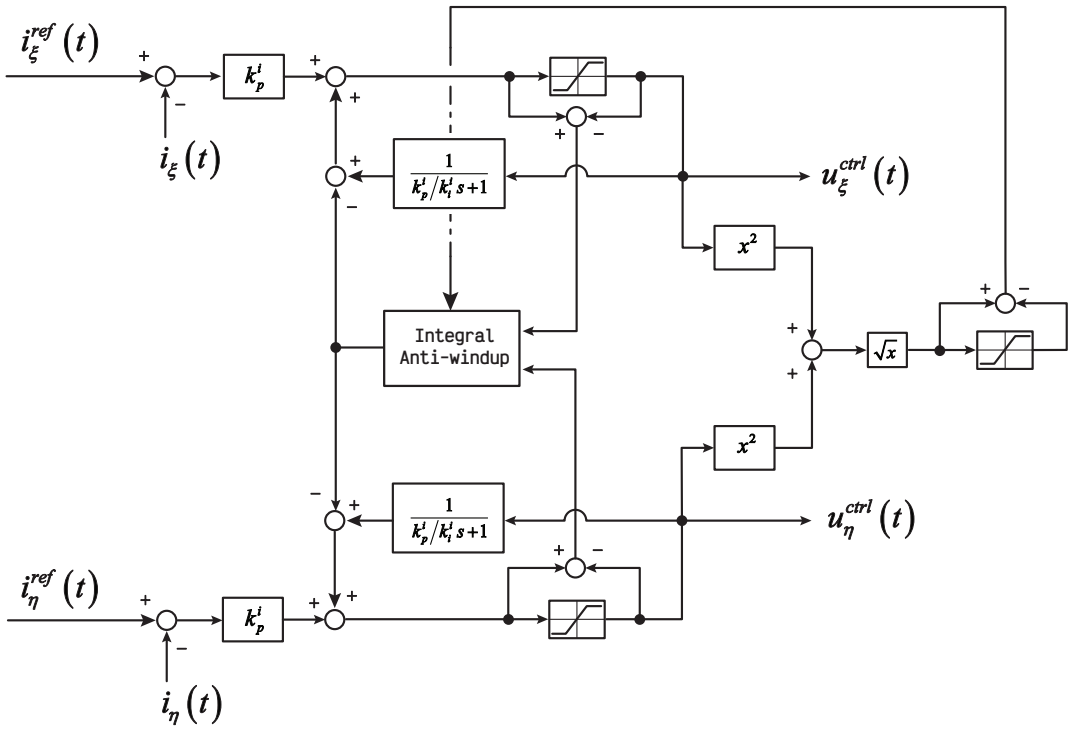


Figure 4: Vector-PI control, where anti-windup is generated by three over-length conditions: on phase D, on phase Q, and on vector amplitude.



4 Simulation Results

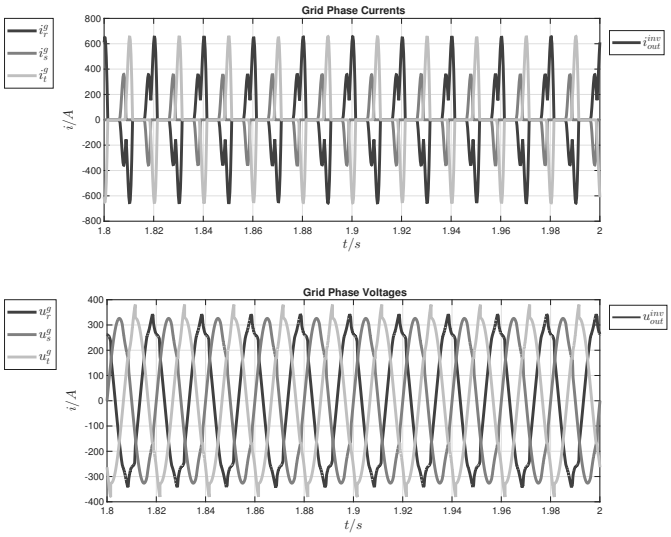
4.1 Output frequency at 50Hz

Quantity	**value**
Load current reference	$i_{out}^{ref} = 600\,\text{A rms}$
Load inductance (at primary side)	$L_{load} = 400\,\mu\text{H}$
Load resistance (at primary side)	$R_{load} = 250\,\text{m}\Omega$
Load frequency	$f_{load} = 50\,\text{Hz}$
Switching frequency (per leg)	$f_{sw} = 10\,\text{kHz}$
DClink voltage	$v_{dc} = 540\,\text{V}$
External gate ON resistance	$R_g^{on} = 0\,\Omega$
External gate OFF resistance	$R_g^{off} = 0\,\Omega$
Turn-On Switching Energy ($R_g^{on} = 0.75\,\Omega$, $T_j = 150\,^\circ\text{C}$, $u_{dc} = 1300\,\text{V}$, $i_d = 1000\,\text{A}$)	$E_{on} = 1.56\,\text{mJ}$
Turn-Off Switching Energy ($R_g^{off} = 1.3\,\Omega$, $T_j = 150\,^\circ\text{C}$, $u_{dc} = 1300\,\text{V}$, $i_d = 1000\,\text{A}$)	$E_{off} = 0.84\,\text{mJ}$
Reserve Recovery Energy ($T_j = 150\,^\circ\text{C}$, $u_{dc} = 1300\,\text{V}$, $i_d = 1000\,\text{A}$)	$E_{rr} = 0.075\,\text{mJ}$
MOSFET Thermal Resistance, Junction to Heatsink (per MOSFET)	$R_{jh} = 25\,^\circ\text{C}\,\text{kW}^{-1}$
Thermal Interface Material (per MOSFET)	$R_{TIM} = 5\,^\circ\text{C}\,\text{kW}^{-1}$
MOSFET Thermal Resistance, Heatsink to Ambient (per MOSFET)	$R_{ha} = 7.5\,^\circ\text{C}\,\text{kW}^{-1}$

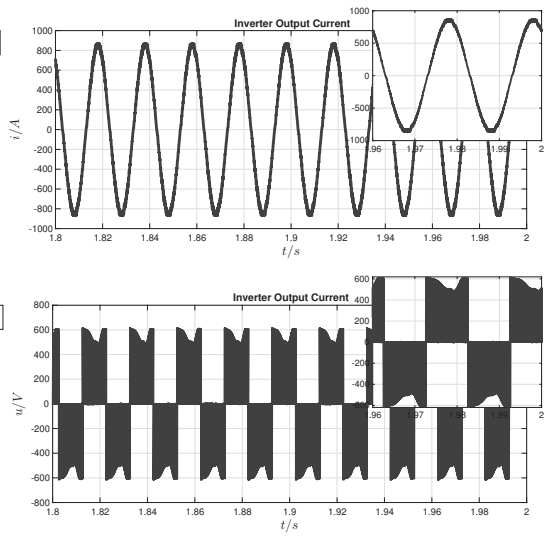
Table 1: List of conditions for the electro-thermal simulation at $f_{load} = 50\,\text{Hz}$.

Quantity	**value**
Load current	$i_{out} = 600\,\text{Arms}$
Load voltage	$u_{out} = 238\,\text{V}$ rms
Grid input active power	$P_g = 93\,\text{kW}$
DClink capaciror current	$i_{ac}^{C} = 343\,\text{Arms}$
MOSFET Q1 rms current	$i_{Q1} = 424\,\text{Arms}$
MOSFET Q1 maximum voltage	$u_{Q1}^{max} = 620\,\text{V}$
MOSFET Q1 power losses	$P_{Q1} = 432\,\text{W}$
Module power losses	$P_m = 864\,\text{W}$
Inverter power losses	$P_H = 1728\,\text{W}$
MOSFET maximum J-H delta temperature	$T_{jh}^{Q1} = 90\,^{\circ}\text{C}$
Diode rectifier rms current	$i_{dr} = 400\,\text{Arms}$

Table 2: List of data results for the electro-thermal simulation at $f_{load} = 50\,\text{Hz}$.

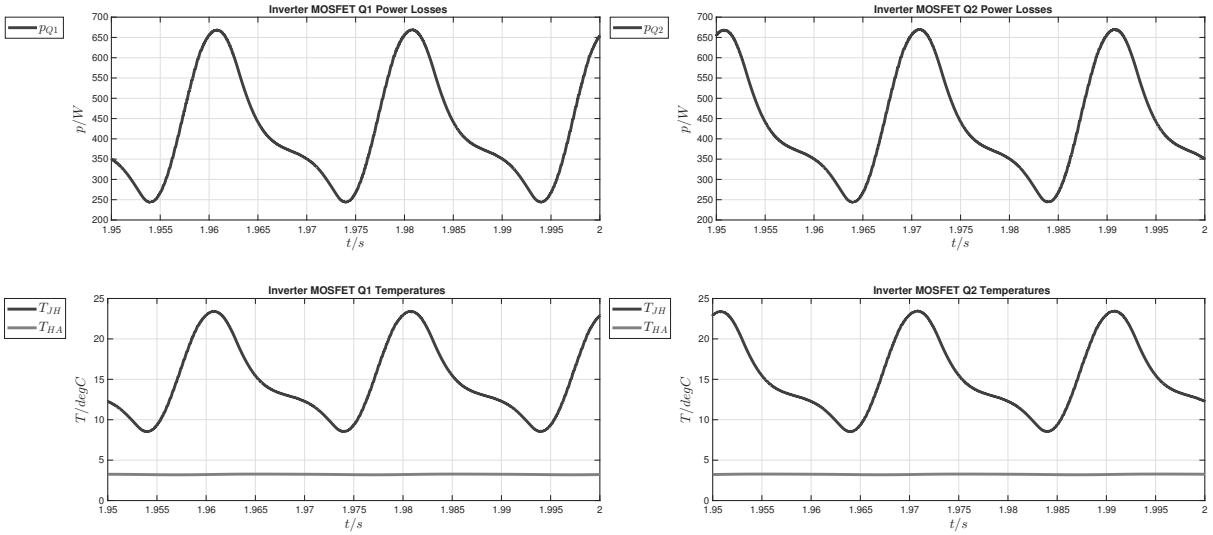


(a) Grid currents (top), and grid voltages (bottom), supposing $S_{nom} = 250\,\text{kV A}$ and $U_{cc} = 4\,\%$.



(b) Inverter output current (top), and inverter output voltage (bottom), supposing $f_{pwm} = 10\,\text{kHz}$ per leg.

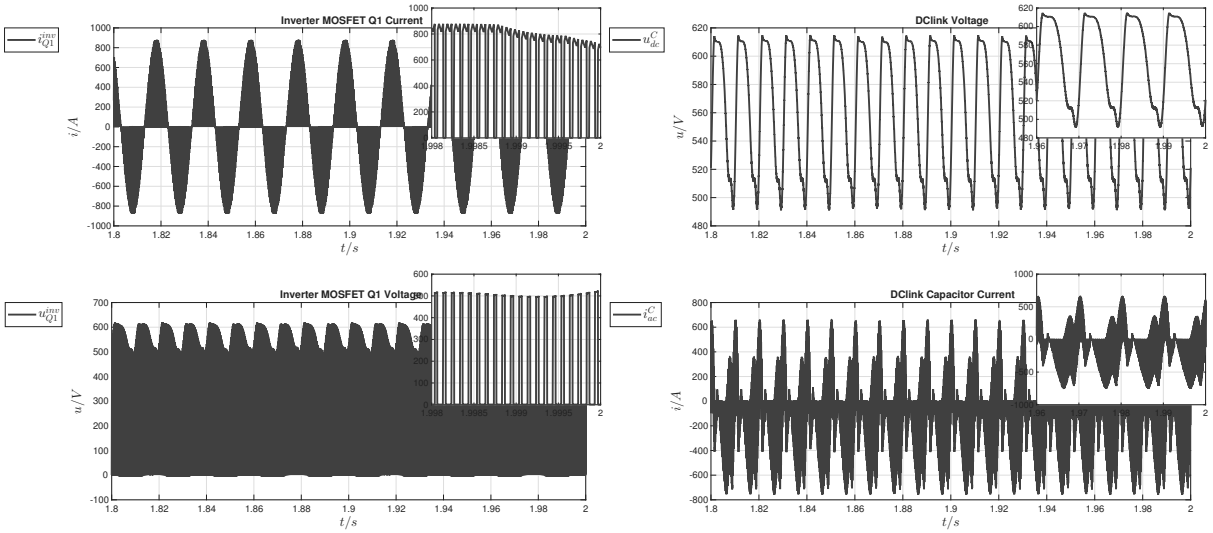
Figure 5: Grid and inverter output quantities.



(a) Inverter MOSFET Q1 power loss (top), and junction-case temperature (bottom).

(b) Inverter MOSFET Q2 power loss (top), and junction-case temperature (bottom).

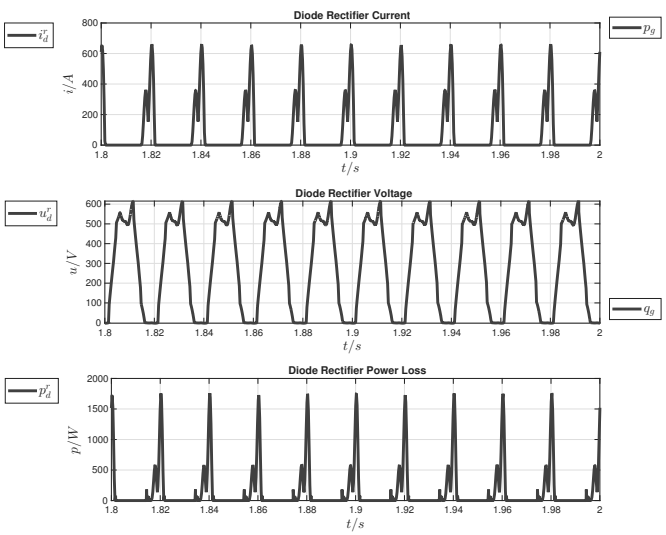
Figure 6: Inverter MOSFETs thermal performance.



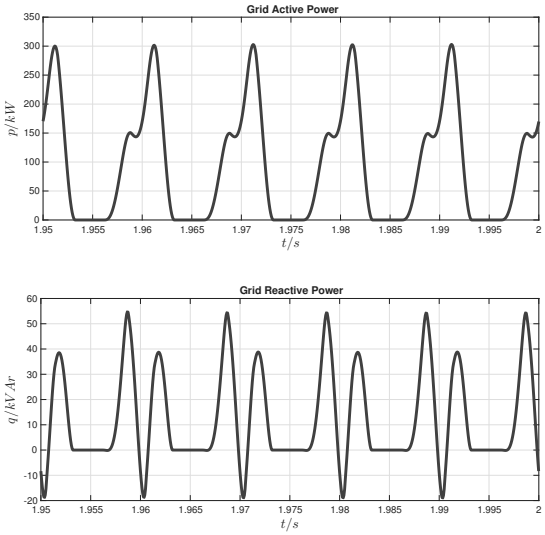
(a) Inverter MOSFET Q1 current (top), and voltage (bottom).

(b) DClink voltage (top), and dclink capcitor current (bottom).

Figure 7: Inverter MOSFETs, and DClink quantities.

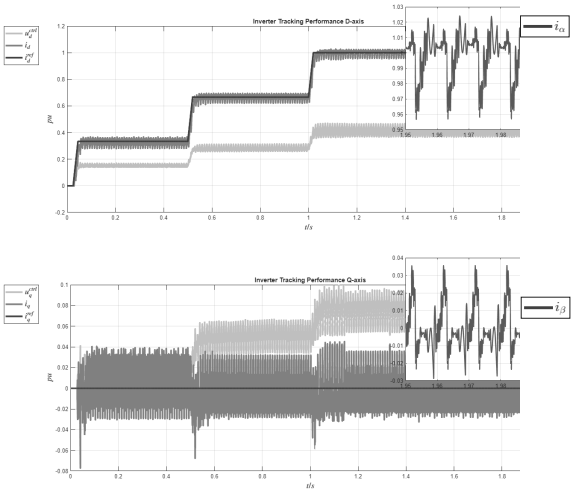


(a) Diode D1 rectifier current (top), voltage (center), and power loss (bottom).

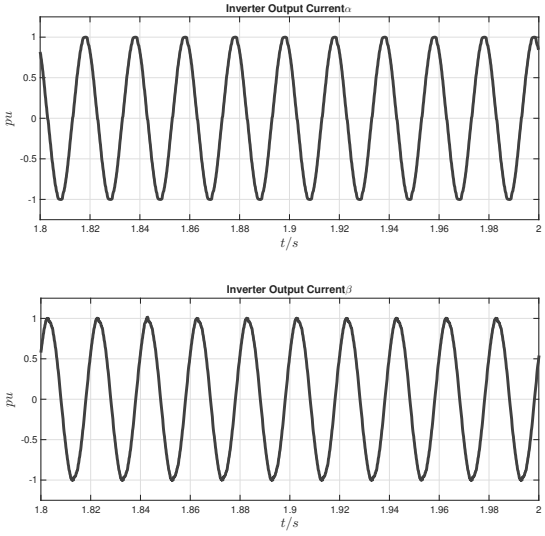


(b) Grid active power (top), and grid reactive power (bottom) - measure implemented according to IEC.

Figure 8: Rectifier diode quantities and grid power analysis.

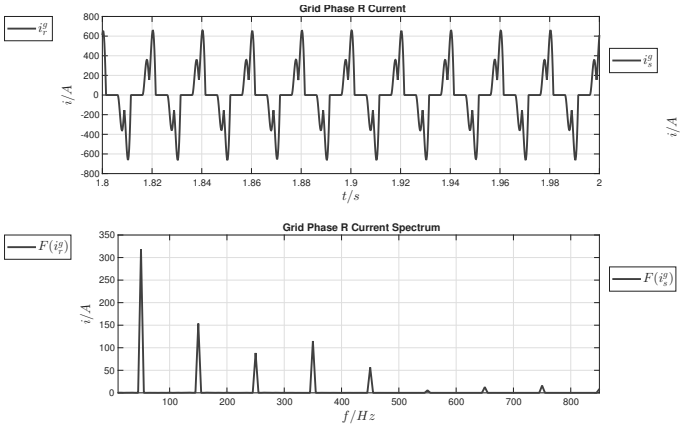


(a) Single phase tracking performance: along D-axis (top), along Q-axis (bottom).

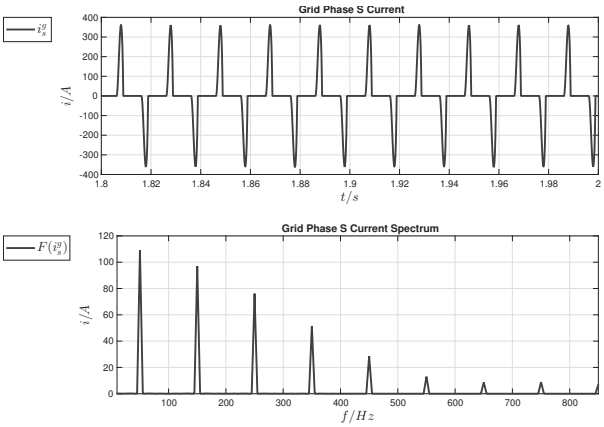


(b) Single phase output current (top), and equivalent shifted by the **phase-shift-filter** (bottom).

Figure 9: Single phase inverter control performance.



(a) Grid phase current R (top), and grid phase current R spectrum (bottom).



(b) Grid phase current S (top), and grid phase current S spectrum (bottom).

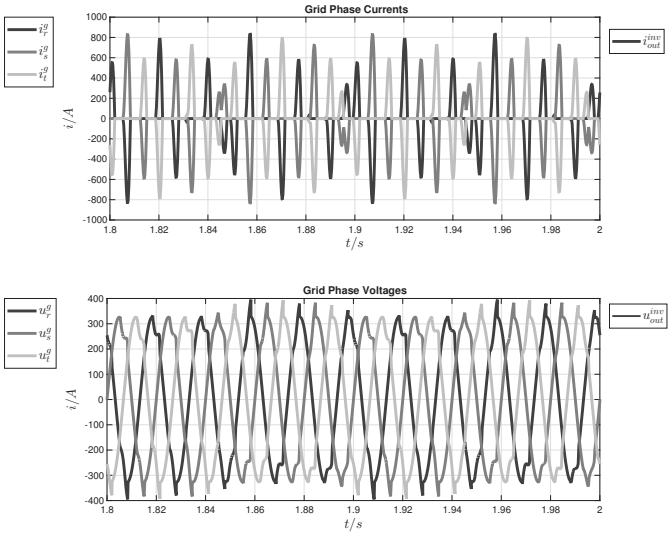
Figure 10: Grid current harmonic analysis.



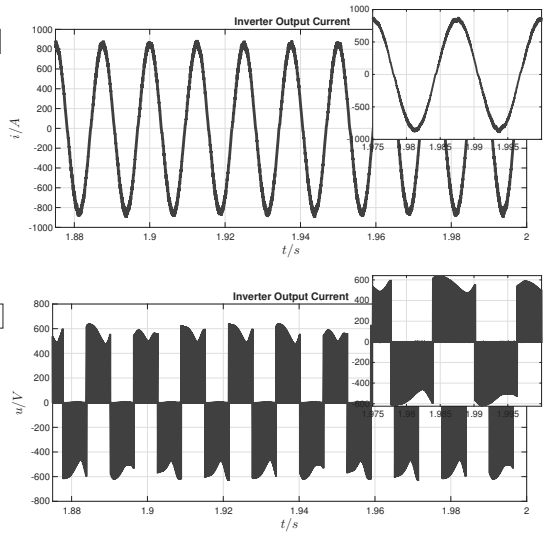
4.2 Output frequency at 80Hz

Quantity	**value**
Load current	$I_m = 45\,\mathrm{A}$ (RMS)
Load voltage	$U_m = 262\,\mathrm{A}$ (ph-ph RMS)
Load power factor	$\cos(\phi) = 0.9$
Load fundamental frequency	$f_{load} = 1250\,\mathrm{Hz}$
Inverter switching frequency	$f_{sw} = 25\,\mathrm{kHz}$
DC-link voltage	$v_{dc} = 400\,\mathrm{V}$
External gate ON resistance	$R_g^{on} = 6\,\Omega$
External gate OFF resistance	$R_g^{off} = 6\,\Omega$
Turn-On Switching Energy ($R_g^{on} = 6\,\Omega$, $T_j = 125\,^\circ\mathrm{C}$, $u_{dc} = 600\,\mathrm{V}$, $i_d = 80\,\mathrm{A}$)	$E_{on} = 1.56\,\mathrm{mJ}$
Turn-Off Switching Energy ($R_g^{off} = 6\,\Omega$, $T_j = 125\,^\circ\mathrm{C}$, $u_{dc} = 600\,\mathrm{V}$, $i_d = 80\,\mathrm{A}$)	$E_{off} = 0.84\,\mathrm{mJ}$
Reserve Recovery Energy ($T_j = 125\,^\circ\mathrm{C}$, $u_{dc} = 600\,\mathrm{V}$, $i_d = 80\,\mathrm{A}$)	$E_{rr} = 0.075\,\mathrm{mJ}$
MOSFET Thermal Resistance, Junction to Heatsink (per MOSFET)	$R_{jh} = 0.642\,^\circ\mathrm{C\,W^{-1}}$
Thermal Interface Material (per MOSFET)	$R_{TIM} = 0.5\,^\circ\mathrm{C\,W^{-1}}$
MOSFET Thermal Resistance, Heatsink to Ambient (per MOSFET)	$R_{ha} = 1.88\,^\circ\mathrm{C\,W^{-1}}$

Table 3: List of condition for the electro-thermal simulation.

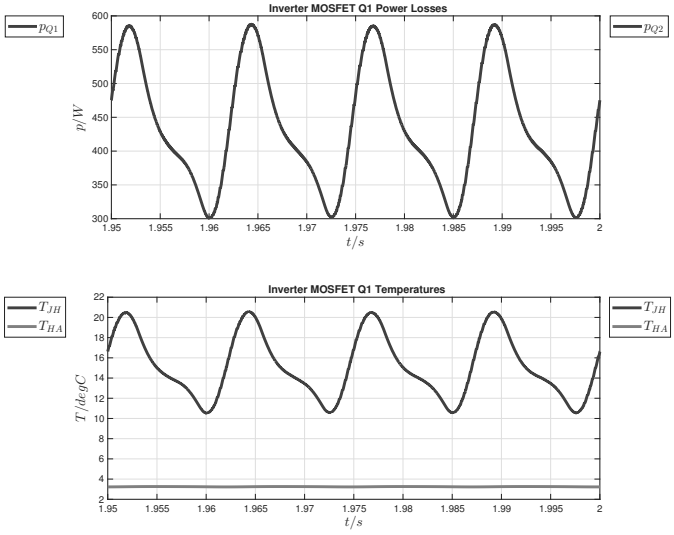


(a) Grid currents (top), and grid voltages (bottom), supposing $S_{nom} = 250\,\text{kV A}$ and $U_{cc} = 4\,\%$.

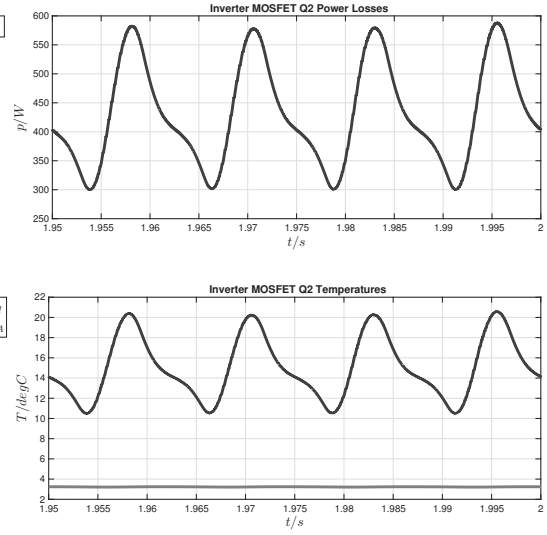


(b) Inverter output current (top), and inverter output voltage (bottom), supposing $f_{pwm} = 10\,\text{kHz}$ per leg.

Figure 11: Grid and inverter output quantities.

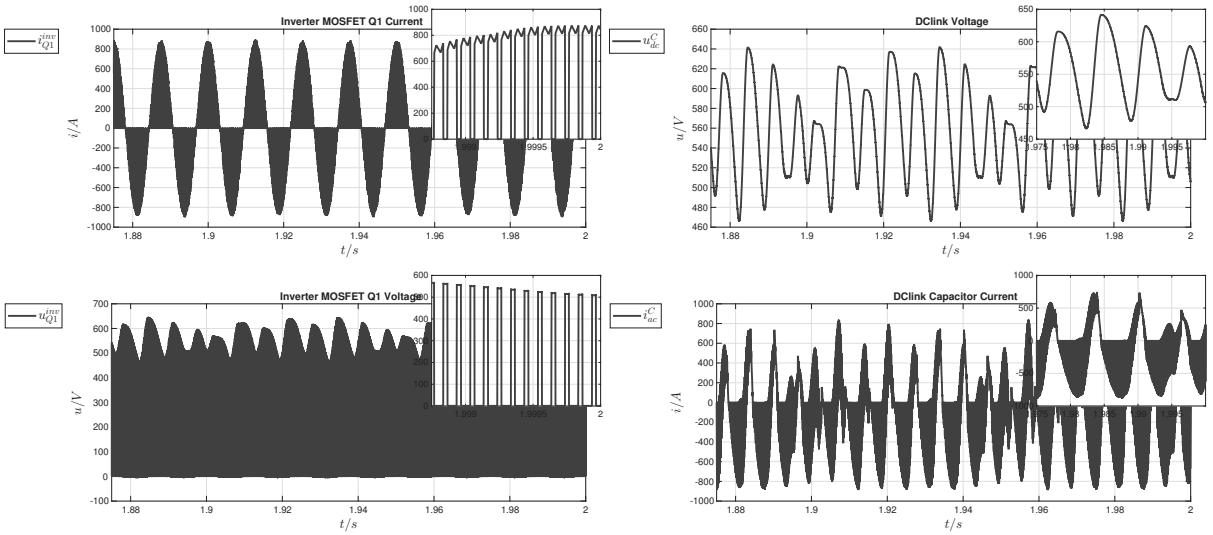


(a) Inverter MOSFET Q1 power loss (top), and junction-case temperature (bottom).



(b) Inverter MOSFET Q2 power loss (top), and junction-case temperature (bottom).

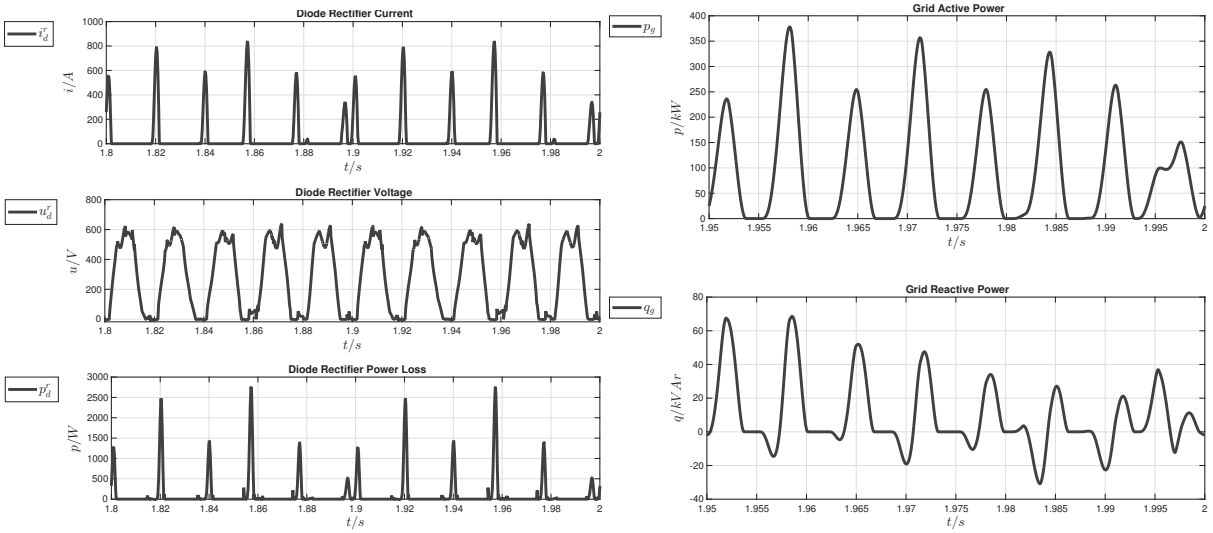
Figure 12: Inverter MOSFETs thermal performance.



(a) Inverter MOSFET Q1 current (top), and voltage (bottom).

(b) DClink voltage (top), and dclink capcitor current (bottom).

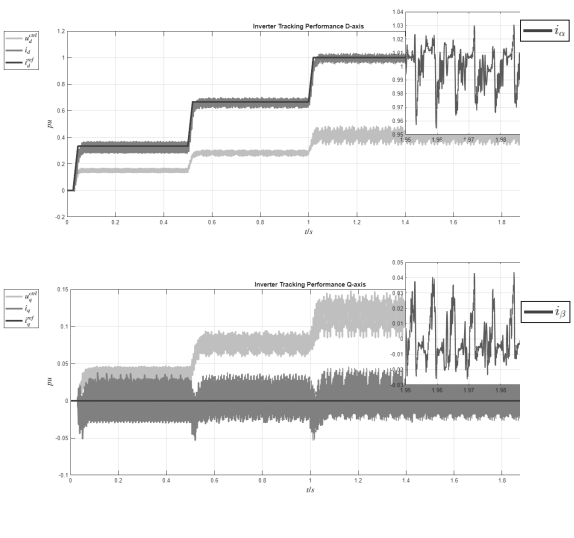
Figure 13: Inverter MOSFETs, and DClink quantities.



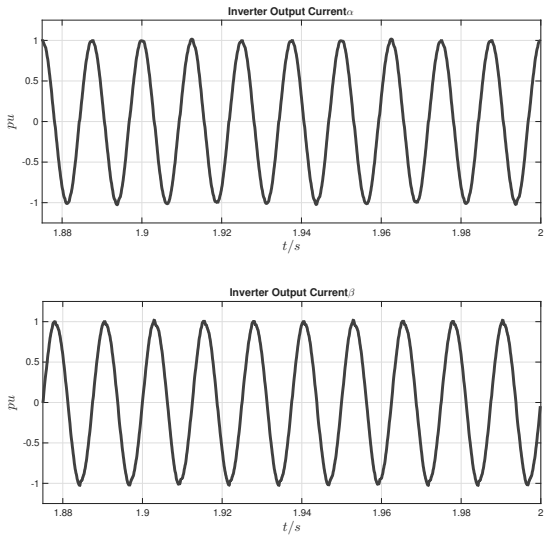
(a) Diode D1 rectifier current (top), voltage (center), and power loss (bottom).

(b) Grid active power (top), and grid reactive power (bottom) - measure implemented according to IEC.

Figure 14: Rectifier diode quantities and grid power analysis.

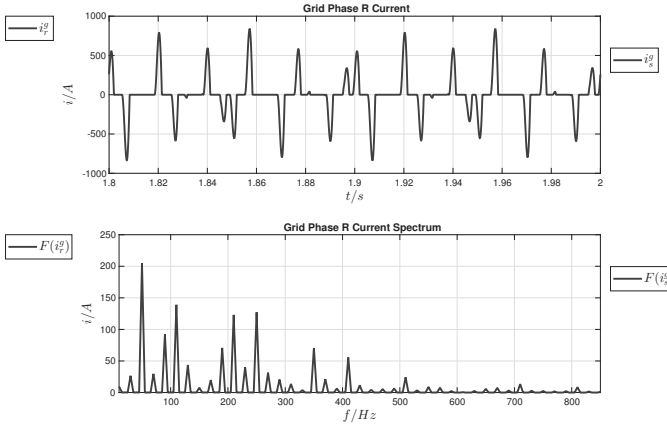


(a) Single phase tracking performance: along D-axis (top), along Q-axis (bottom).

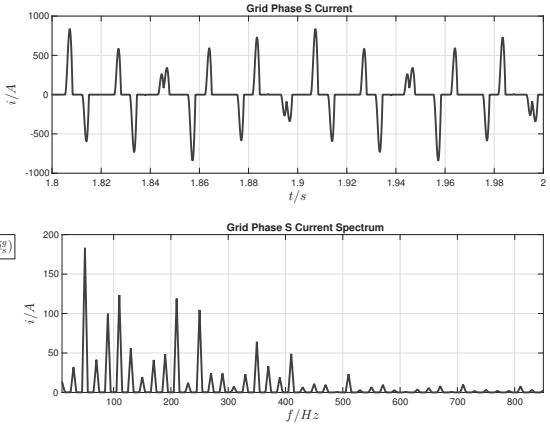


(b) Single phase output current (top), and equivalent shifted by the **phase-shift-filter** (bottom).

Figure 15: Single phase inverter control performance.



(a) Grid phase current R (top), and grid phase current R spectrum (bottom).



(b) Grid phase current S (top), and grid phase current S spectrum (bottom).

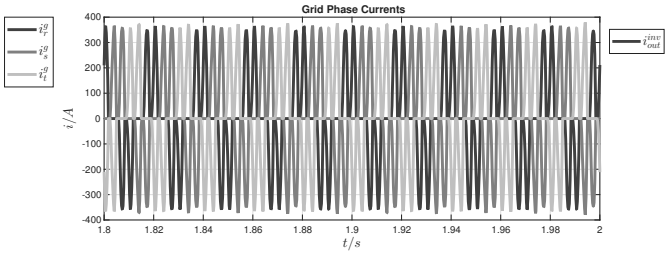
Figure 16: Grid current harmonic analysis.



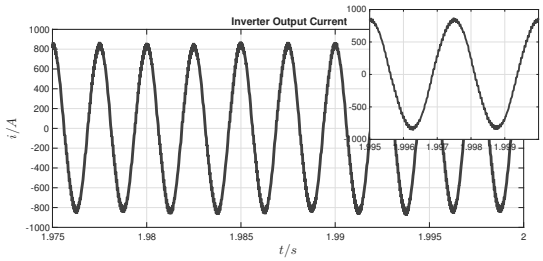
4.3 Output frequency at 400Hz

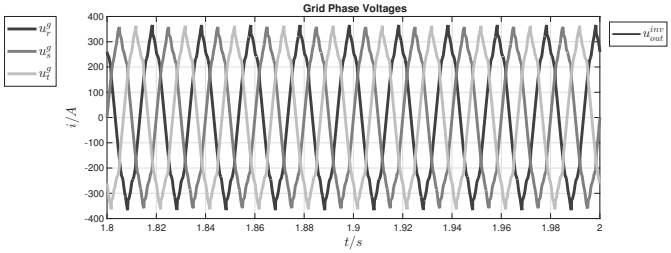
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Load power factor	$\cos(\phi) = 0.9$
Load fundamental frequency	$f_{load} = 1250\,\mathrm{Hz}$
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External gate ON resistance	$R_g^{on} = 6\,\Omega$
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Turn-On Switching Energy ($R_g^{on} = 6\,\Omega$, $T_j = 125\,^\circ\mathrm{C}$, $u_{dc} = 600\,\mathrm{V}$, $i_d = 80\,\mathrm{A}$)	$E_{on} = 1.56\,\mathrm{mJ}$
Turn-Off Switching Energy ($R_g^{off} = 6\,\Omega$, $T_j = 125\,^\circ\mathrm{C}$, $u_{dc} = 600\,\mathrm{V}$, $i_d = 80\,\mathrm{A}$)	$E_{off} = 0.84\,\mathrm{mJ}$
Reserve Recovery Energy ($T_j = 125\,^\circ\mathrm{C}$, $u_{dc} = 600\,\mathrm{V}$, $i_d = 80\,\mathrm{A}$)	$E_{rr} = 0.075\,\mathrm{mJ}$
MOSFET Thermal Resistance, Junction to Heatsink (per MOSFET)	$R_{jh} = 0.642\,^\circ\mathrm{C\,W^{-1}}$
Thermal Interface Material (per MOSFET)	$R_{TIM} = 0.5\,^\circ\mathrm{C\,W^{-1}}$
MOSFET Thermal Resistance, Heatsink to Ambient (per MOSFET)	$R_{ha} = 1.88\,^\circ\mathrm{C\,W^{-1}}$

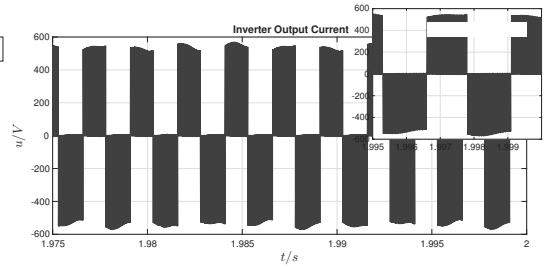
Table 4: List of condition for the electro-thermal simulation.



(a) Grid currents (top), and grid voltages (bottom), supposing $S_{nom} = 250\,\text{kV A}$ and $U_{cc} = 4\,\%$.

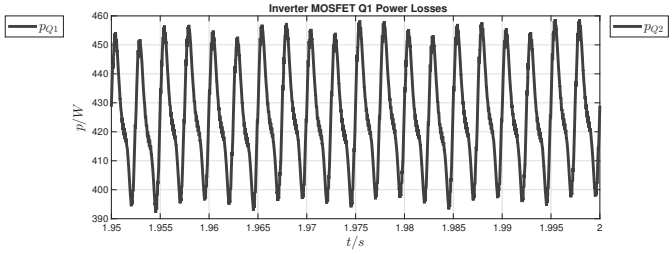


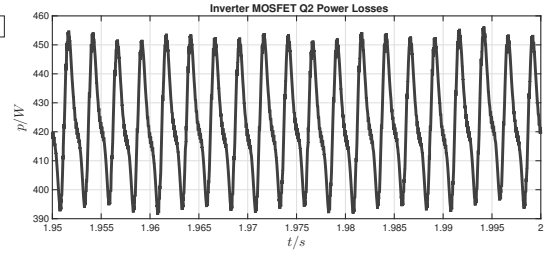


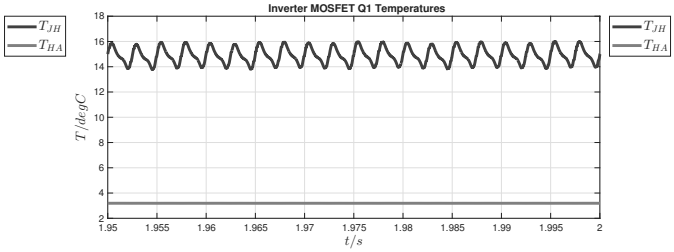


(b) Inverter output current (top), and inverter output voltage (bottom), supposing $f_{pwm} = 10\,\text{kHz}$ per leg.

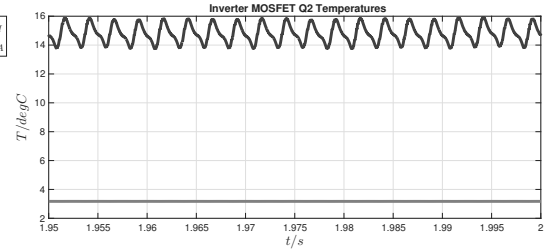
Figure 17: Grid and inverter output quantities.





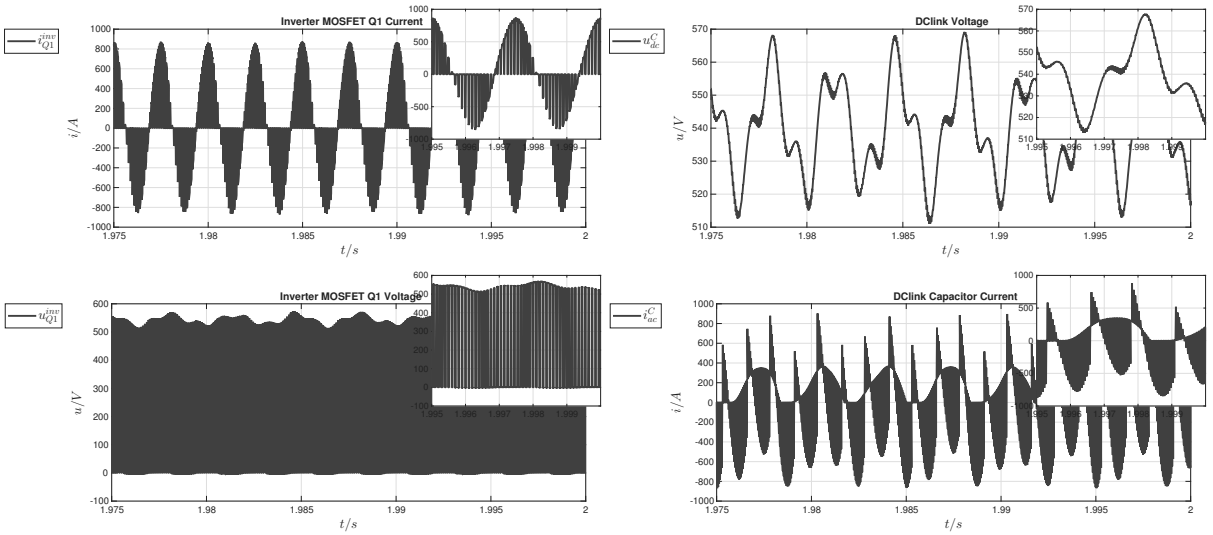


(a) Inverter MOSFET Q1 power loss (top), and junction-case temperature (bottom).



(b) Inverter MOSFET Q2 power loss (top), and junction-case temperature (bottom).

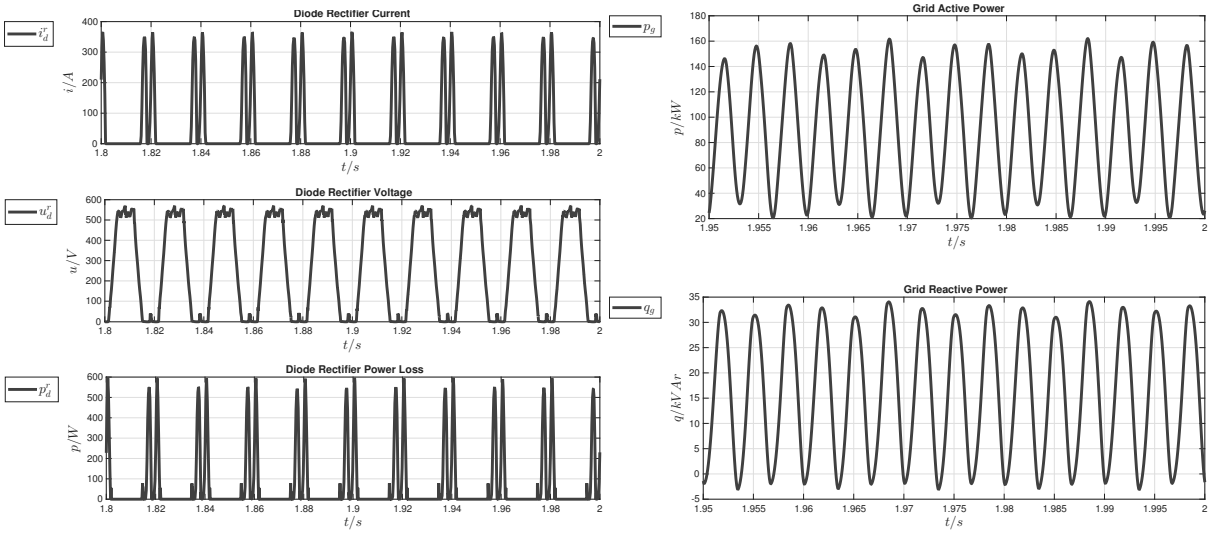
Figure 18: Inverter MOSFETs thermal performance.



(a) Inverter MOSFET Q1 current (top), and voltage (bottom).

(b) DClink voltage (top), and dclink capcitor current (bottom).

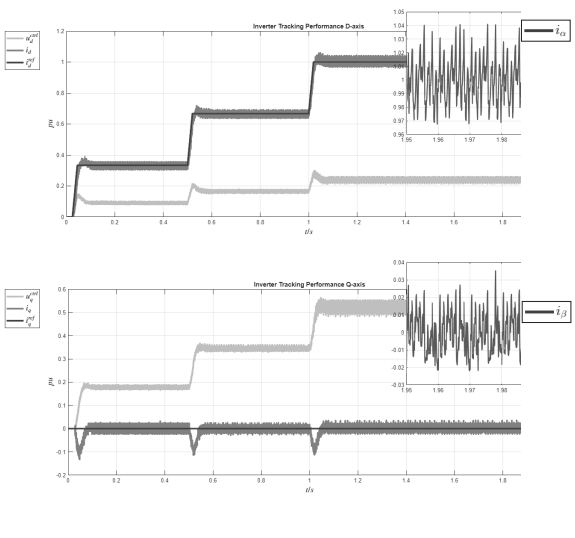
Figure 19: Inverter MOSFETs, and DClink quantities.



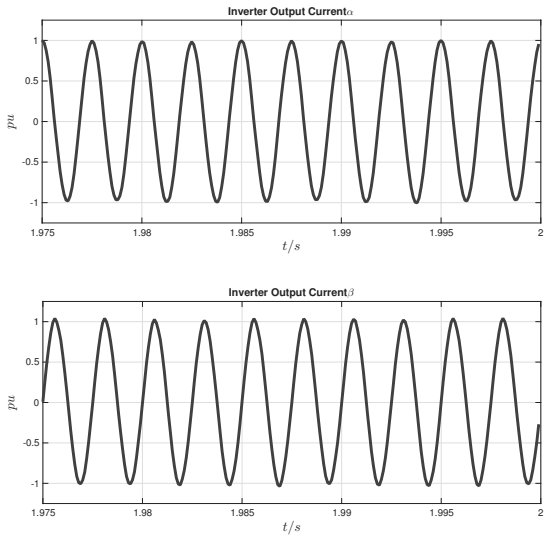
(a) Diode D1 rectifier current (top), voltage (center), and power loss (bottom).

(b) Grid active power (top), and grid reactive power (bottom) - measure implemented according to IEC.

Figure 20: Rectifier diode quantities and grid power analysis.

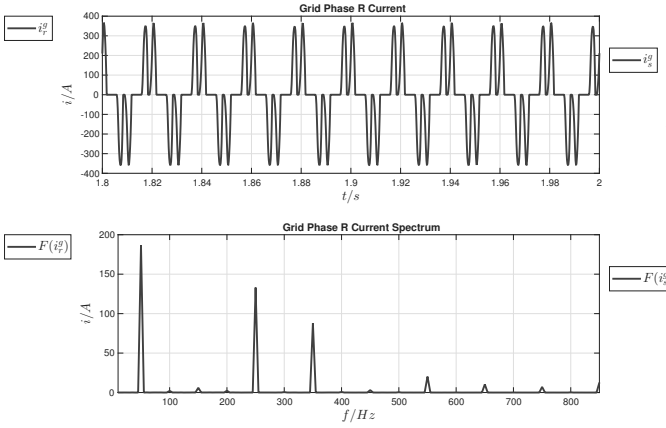


(a) Single phase tracking performance: along D-axis (top), along Q-axis (bottom).

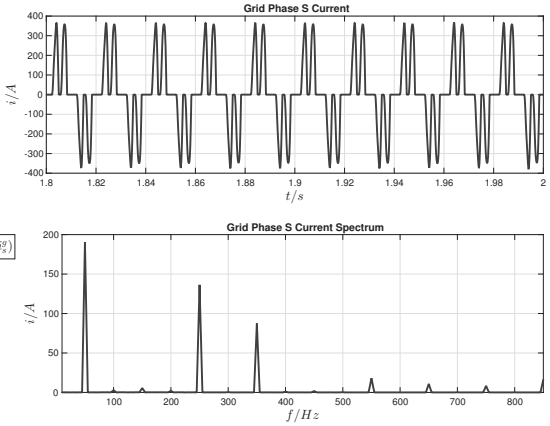


(b) Single phase output current (top), and equivalent shifted by the **phase-shift-filter** (bottom).

Figure 21: Single phase inverter control performance.



(a) Grid phase current R (top), and grid phase current R spectrum (bottom).



(b) Grid phase current S (top), and grid phase current S spectrum (bottom).

Figure 22: Grid current harmonic analysis.

References

[1] R. Teodorescu, M. Liserre, P. Rodriguez, *Grid converters for photovoltaic and wind power systems*. Wiley, 2011.